

- 2 (a) The house mouse, *Mus musculus*, has a diploid number of 40 chromosomes.

Fig. 2.1 shows 6 of these chromosomes.

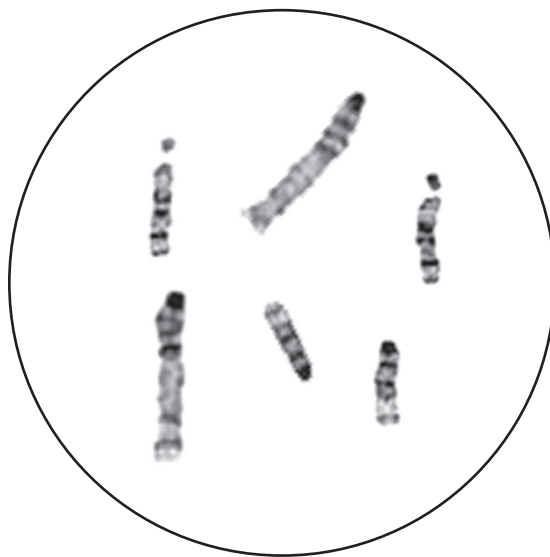


Fig. 2.1

Identify **one** pair of homologous chromosomes on Fig. 2.1 by drawing circles around **two** chromosomes. [1]

- (b) Fig. 2.2 shows the banding pattern of chromosome pair 11 of *M. musculus*. The banding pattern is obtained by staining.

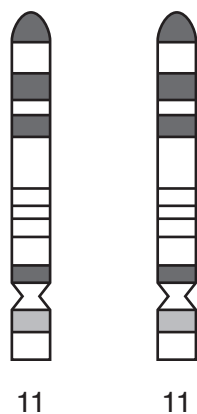


Fig. 2.2

- (i) Explain why chromosomes, such as those in Fig. 2.2, are described as a homologous pair.

.....

.....

.....

.....

.....

..... [3]

- (ii) State the number of chromosomes that are present in *M. musculus* spermatozoa.

..... [1]

- (c) *M. musculus* produces gametes by meiosis. These gametes are genetically different.

There is random fusion of gametes at fertilisation.

- (i) Explain why meiosis is important in the life cycle of *M. musculus*, **apart from** producing genetically different gametes.

.....

.....

.....

..... [2]

- (ii) Explain how the random fusion of gametes leads to the expression of rare, recessive alleles.

.....

.....

.....

.....

..... [2]

(d) A mutation causing coats of mice to be woolly in appearance is in a gene located on chromosome 11. The mutation causes a very shortened polypeptide product. Mice with the woolly coat phenotype have longer fur than mice with normal coats.

(i) Explain how a base substitution mutation can lead to a very shortened polypeptide product.

.....

.....

.....

.....

.....

..... [2]

(ii) The inheritance of the woolly coat characteristic was investigated.

Draw a genetic diagram to show a cross between two heterozygous parents with normal coats.

Use the symbols **A** and **a** for the alleles.

parental genotypes

gametes

offspring genotypes

offspring phenotypes

[3]

[Total: 14]